

Re-evaluating Inherited SIF Anomaly Detection Methods: A 3D Convolutional Autoencoder Approach

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Key Points:

- We calculate TROPOMI SIF anomalies using convolutional autoencoders (CAE) and standard deviations (z-scores) for the Amazon in 2024.
- Anomalies derived from CAE show strong correlation with national drought monitor data and drought index SPEI over the Amazon.
- Calculated anomalies are highly sensitive to the chosen methodology and careful method selection is thus advised.

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Abstract

In the past decade, the Amazon has experienced multiple severe droughts, raising critical questions about how vegetation stress can be detected using remote sensing indicators such as Solar-Induced Chlorophyll Fluorescence (SIF). In this study, we compare two anomaly detection approaches applied to TROPISIF data. First a classical pixel wise z-score method, and second a 3D Convolutional Autoencoder (CAE) that learns spatiotemporal SIF patterns. We evaluate both methods against two drought products, the global SPEI index and Brazil’s Monitor de Secas (MSB). Our results show that the CAE identifies broader, more persistent anomalies than the z-score method and aligns more closely with drought patterns detected by SPEI and MSB, particularly during the peak dry period in September 2024 and the months of delayed recovery that followed. These findings highlight how methodological choices shape the interpretation of vegetation stress and why inherited methodologies must be challenged.

Plain Language Summary

Researchers use satellite images to study regions around the world. In particular, they are interested in anomalous events, such as droughts or heatwaves, because these often have large impacts on ecosystems. To find locations where vegetation behaves differently than normal, anomaly detection methods can be used. Here we compared a simple classical method based on the standard deviation of the satellite measurement at a given location, with a more comprehensive method that considers both variations in time and space. We find that the choice of method has quite a large influence on the locations that are identified as being anomalous. Therefore, we stress that it is important that researchers are aware of this when selecting an anomaly detection method.

1 Introduction

Extreme droughts are becoming more frequent in the Amazon, reducing the carbon uptake capability and threatening to accelerate the processes of forest degradation (Chambers et al., 2026; Flores et al., 2024; Gatti et al., 2021; Van Passel et al., 2024). Recent studies using local forest measurements (Brienen et al., 2015; Hubau et al., 2020) and top-down approaches (Botía et al., 2025; Rosan et al., 2024) have shown that the uptake of carbon by the Amazon is decreasing. This long-term change and the increased occurrence of drought also affect the hydrological cycle (Fleischmann et al., 2025; Staal

et al., 2023; van Schaik et al., 2018) and fire dynamics (Aragão et al., 2018; Naus et al., 2022) of the ecosystem. Monitoring all these processes is crucial for anticipating how future climatic events may unfold (Marengo et al., 2018; Smallman et al., 2021) and for designing effective strategies to prevent or mitigate ecosystem collapse.

Remote sensing provides a valuable tool for studying ecosystems, as it allows us to monitor vast and difficult-to-access regions (Kooistra et al., 2024). Among the available signals, Solar-Induced Chlorophyll Fluorescence (SIF) has emerged as a promising indicator of vegetation productivity. SIF is a faint light re-emitted by chlorophyll during photosynthesis, with approximately 1–2% of absorbed sunlight released as fluorescence (Frankenberg & Berry, 2018; Y. Zhang & Zhang, 2026). SIF was used to study vegetation activity across different ecosystems, focusing on seasonal dynamics (Mengistu et al., 2021; Turner et al., 2020), drought impacts (Buitink et al., 2020; Jiang et al., 2025; Smith et al., 2020) and ecosystem recovery after disturbances (Anema et al., 2024; Lu et al., 2024).

SIF has also been used to study ecosystem dynamics in the Amazon (Doughty et al., 2019; Lee et al., 2013; Z. Zhang et al., 2023) and SIF anomalies have already proven useful in studying the impacts of drought on photosynthesis in the Amazon (Botía et al., 2025; Koren et al., 2018; Liu et al., 2017). When plants experience stress, we expect extreme drought to coincide with reduced SIF, as photosynthetic capacity declines. However, the relationship between meteorological drought indices and vegetation stress is not straightforward: while meteorological indices may indicate drought conditions, plants may rely on deeper water reserves or adjust physiologically, leading to a weaker or delayed SIF response. Thus, the question arises: can we gain more interpretation of SIF signals in relation to extreme droughts using different concepts of “anomaly”?

Traditionally, anomalies in univariate datasets such as SIF have been studied using statistical approaches like the z-score. These methods are simple, interpretable, and computationally efficient, but they fail to capture complex spatio-temporal dependencies as each grid cell is treated as an independent entity. With the rise of artificial intelligence in the 21st century, machine learning methods have increasingly been applied to Earth Observation datasets (Sun et al., 2024; Zhu et al., 2017), including the development of Foundation Models such as TerraMind (Jakubik et al., 2025) and AlphaEarth Foundations (Brown et al., 2025). That is why, introducing machine learning for the task

of anomaly detection of SIF could be to our advantage. Convolutional Autoencoders (CAEs), in particular, are well-suited for SIF data, as they can learn both seasonal cycles and spatial patterns characteristic of the Amazon basin, an area of difficulty for research. Nevertheless, such methods have not yet been used for detecting anomalies in biosphere signals, such as SIF, and are often criticized for their “black-box” nature, which limits interpretability.

In this study, we investigate how SIF anomalies detected with a classical z-score approach compare with those obtained using a CAE on the TROPOSIF dataset (Guanter et al., 2021), a SIF product retrieved from the TROPOMI sensor on the Sentinel-5p satellite (Veefkind et al., 2012). Our analysis focuses on the Amazon rainforest during 2024, a year marked by extreme drought conditions (Botía et al., 2025; Marengo et al., 2026; Toreti et al., 2024). Specifically, we ask:

- What differences arise between the anomalies detected by the z-score method and those identified by the CAE?
- How are anomalies in vegetation related to drought conditions, in particular during the onset and recovery of drought?

To assess this, we combine two complementary drought products: the Monitor de Secas (MSB), which provides qualitative and regionally validated drought assessments, and the Standardized Precipitation-Evapotranspiration Index (SPEI), a globally consistent quantitative index. By comparing these with the anomaly signals from both approaches, we aim not to validate one method over the other, but rather to shed light on what the CAE can contribute beyond the classical z-score in understanding vegetation stress and resilience under extreme drought.

2 Data and Methods

2.1 Sun-Induced Fluorescence Dataset

We use SIF from the TROPOSIF dataset (Guanter et al., 2021) developed from optical remote sensing data, we take monthly SIF data from 2019-2024 (six years) L3-type data. This dataset combines a global continuous spatial sampling with a 3.5×7.5 km² pixel size at nadir in the near infrared (3.5×5.5 km² since August 2019). We focus on the northern Amazonia region including parts of Brazil, Peru, Colombia, and other

105 countries, bounded by latitudes 15°S to 5°N along longitudes 75°W to 50°W (see Fig.
106 1).

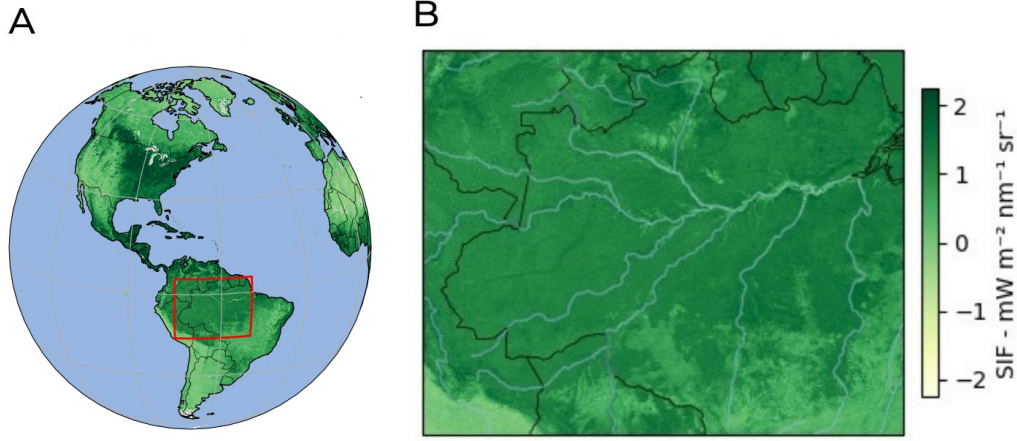


Figure 1: Spatial patterns of TROPOMI Solar-Induced Chlorophyll Fluorescence (SIF). (A) Projection for July 2020 and indication of study area (red box) bounded between latitudes 15°S to 5°N and longitudes 75°W to 50°W. (B) Climatology for July on zoomed-in view of the study area.

107 We are interested in anomalies on the behavior of SIF over the Amazonia both tem-
108 porally and spatially. From the TROPOSIF dataset we have access to five cubes for the
109 reference period (2019-2023, so excluding the year 2024) of SIF data where the three di-
110 mensions are months, latitude and longitude. We performed an augmentation method
111 that increases the data size, from five to 500 datapoints for increased robustness of the
112 statistical analysis. First we spatially cropped the data to dimensions divisible by 10 (910
113 $\times 1130$) by removing some pixels. Then we created 100 distinct low-resolution (91×113)
114 versions by systematically sampling every 10th pixel with different offsets. This resulted
115 in an augmented dataset of shape (500, 12, 91, 113). This approach provides cubes with
116 less spatial resolution while maintaining key geographical features (see Fig. S1 in the Sup-
117 porting Information) and providing sufficient data for detecting anomalies with both the
118 CAE and the z-score method.

2.2 Anomaly Detection

2.2.1 Z-score Method

The z-score method identifies anomalies by quantifying how far individual data points deviate from the mean, measured in units of standard deviation. This approach is widely used in anomaly detection for its simplicity and effectiveness, especially in time series and satellite data, as it does not require prior knowledge of anomaly timing and can be adapted to dynamic baselines. For each pixel and month, we compute the mean and standard deviation for the years 2019-2023. For the year 2024, the z-score is calculated as

$$z_{x,y,t} = (\text{value}_{x,y,t} - \text{mean}_{x,y,t}) / \text{std}_{x,y,t} \quad (1)$$

where x and y refer to the spatial location of the pixel and where t refers to the month. For each pixel there will thus be 12 z-scores for the year 2024. An anomaly is flagged when the absolute z-score exceeds 2, i.e. $|z| > 2$. This statistical method provides a straightforward baseline for identifying spatial and temporal outliers by measuring how far a value deviates from historical norms.

2.2.2 CAE Method

We built a 3D Convolutional Autoencoder (CAE) trained on the 500 datacubes of augmented data based on years 2019–2023 of SIF data, with 20% of the data reserved for validation to ensure a robust assessment of generalization performance. The model processes input cubes of shape $(12 \times 91 \times 113)$, corresponding to one year of monthly SIF data across the study region. The CAE consists of:

1. Encoder: three 3D convolutional layers (ReLU activation) with max-pooling, progressively reducing dimensionality.
2. Bottleneck: compressed latent representation $(3 \times 23 \times 29 \times 8)$.
3. Decoder: symmetric upsampling and convolutional layers to reconstruct inputs.
4. Output: single-channel reconstruction cropped to match input dimensions.

A visualization is provided in Fig. S2 of the Supporting Information.

The network learns the baseline climatology by minimizing the reconstruction error from the input data: the difference between original and reconstructed data (see Fig.

S3 in the Supporting Information for more details). For our 3D SIF data cubes, we use three key layers. The first one is 3D Convolutions: where kernel filters are convoluted (taking the inner product with the kernel and input) through the data to detect local spatiotemporal patterns. The second one is Max-pooling: which simplifies the data by keeping the maximum value in each small block covering the cube. The last one is Up-sampling: which expands the compressed data back to full resolution by repeating input values.

The model has 9,553 parameters (see Table S1 in the Supporting Information for more details) and a batch size of 30 which balances gradient stability and memory use. We use ReLU activations for their computation efficiency, non-linearity and because this avoids vanishing gradients for positive activations. We used the Adam optimizer for its adaptive learning rate capabilities, which facilitate effective convergence. The mean squared error (MSE) loss was employed as it is well-suited for reconstruction tasks in autoencoder training. Training was conducted over 50 epochs, a choice informed by monitoring of loss curves and to preserve the model’s sensitivity to anomalies. The model achieved a final training loss of 0.0695 and a validation loss of 0.0735, indicating good generalization without overfitting which validates our architecture’s big parameter size.

We observe that a significant portion of the latent space is unused. Meaning that the model’s capacity to memorize the training data, including anomalies, is limited. This acts as a strong form of regularization, forcing the autoencoder to focus on reconstructing only the most dominant, consistent patterns in the data, and ignore rare or inconsistent patterns. The reconstructed images retain only the principal spatial and temporal features of the input data. This level of reconstruction fidelity is desirable in our context, as it prevents the model from overfitting to specific details such as potential anomalies present in the training set. This is an additional help for the small amount of data available.

The CAE is trained to minimize reconstruction error on the baseline years (2019–2023). After training, normal samples should reconstruct with low error, while anomalous ones yield higher error. To define anomalies, we compute pixel-wise monthly reconstruction errors over the baseline period and select the 99th percentile as a threshold. The model goes through 2024 data and we mark a pixel anomalous if its reconstruction error exceeds this threshold. The anomaly detection strategy leverages the trained CAE to iden-

tify deviations from typical spatial and temporal patterns in SIF data. This method allows for robust detection of outliers while minimizing false positives in areas with naturally low or high SIF, thereby providing a more context-aware characterization of anomalous behavior in the dataset.

2.3 Drought Datasets and Analysis

To assess the relationship between SIF anomalies and drought conditions, we use two complementary drought products.

- The Global Standardized Precipitation Evapotranspiration Index (SPEI) database (SPEIbase) provides long-term, robust information about drought at the global scale with 0.5° spatial resolution and monthly time resolution (Beguería et al., 2010; Vicente-Serrano et al., 2010). It is based on precipitation and potential evapotranspiration (PET) from the Climatic Research Unit (University of East Anglia) and extends back to January 1901. PET is estimated using the FAO-56 Penman-Monteith method, considered reliable for long-term climatological studies. The dataset consists of standardized values across land pixels. SPEIbase offers a quantitative, globally consistent index of drought. We classified pixels with $|\text{SPEI}| > 1$ as anomalous (with negative values indicating dry conditions and positive values corresponding to wet conditions).
- The Monitor de Secas do Brasil (MSB) is a regional drought monitoring product (Agência Nacional de Águas e Saneamento Básico, 2025). It integrates multiple sources; such as precipitation, soil moisture, vegetation indices, and hydrological observations, and is reviewed by local experts before the final drought map is released. The MSB provides a qualitative but locally informed assessment that reflects regional knowledge and context. MSB distinguishes between five different drought severities, ranging from weak drought (‘Seca Fraca S0’) to exceptional drought (‘Seca Excepcional S5’).

Comparing SIF anomalies with both datasets allows us to capture two perspectives: a global, quantitative drought index (SPEIbase) and a regional monitoring product (MSB) driven by local experts. We compute overlap between SIF anomalies and the drought dataset anomalies using the Dice coefficient (Dice, 1945) and the Intersection over Union

(IoU), treating anomalous pixels as binary masks. The Dice coefficient measures the similarity between two sets as twice their intersection divided by their combined size, while the IoU quantifies the ratio of the intersection to the union of the two sets.

3 Results and Discussion

3.1 Comparison between CAE and Z-score Anomalies

Overall, the CAE detects a higher proportion of anomalies compared to the z-score approach as seen in Table 1. Both methods identify September as the month with the highest anomaly prevalence, with 46.69% (CAE) and 35.81% (z-scores). In contrast, the months with the lowest anomaly percentages differ: April for the CAE (16.18%) and July for the z-scores (0.61%). Temporal variability also diverges between the two methods. The z-scores show abrupt changes, most notably from July to August (+15.66%pt) and September to October (−32.07%pt). The CAE exhibits smoother month to month transitions (+9.41%pt and −8.74%pt over the same periods).

To better understand the nature of the detected anomalies, we analyse them in a positive–negative framework. For the z-score method (Figure 2), anomalous values above +2 are shown in on a red scale (where the color refers to the 2024 anomaly, i.e., the 2024 SIF value minus the SIF climatology for that corresponding month), and those below −2 on a blue scale (a discrete version of this is provided as Fig. S4 in the Supporting Information). This visualization highlights diagonal stripe patterns, most visible in January. These spatially rigid, scanline like features over the Amazon are consistent with satellite artifacts rather than real biophysical anomalies.

For the CAE method, all non-anomalous pixels are masked, and anomalous pixels are displayed as the difference between the original observation and its reconstruction with the CAE method. For a positive value, the model expected less SIF and for the negative values the model expected more SIF (see Figure 2). This approach reveals patterns that differ substantially from the z-score output. The diagonal striping characteristic of sensor noise nearly disappears. Notably, the strong artifact signature observed in January in the z-scores does not manifest in the CAE results. Instead, the CAE highlights spatially coherent regions of reduced SIF that correspond more closely to known environmental conditions in early 2024, including lingering El Niño impacts from 2023 (Botía et al., 2025; Espinoza et al., 2024). The CAE also captures a realistic recovery

across parts of the basin during May, consistent with the onset of the rainy season and hot and dry periods later in the year 2024 (Marengo et al., 2026; Toreti et al., 2024).

A particularly interesting contrast between the two methods emerges in the late 2024 period. Both detect strong anomalies in August and September, but the z-score method shows a rapid return to normal conditions in October. This abrupt recovery appears inconsistent with the state of the Amazonas at that time which is discussed further in Sections 3.2 and 3.3. The CAE, on the other hand, reveals a gradual transition, with anomalies weakening in October and November before intensifying again in December.

In Table 1 we compare the percentage of anomalies detected by both methods across 2024. Both approaches identify September as the peak anomaly month, coinciding with the most extreme drought conditions measured by Monitor de Secas and the SPEIbase dataset. The z-score method detects lower anomalies than the CAE method across the year, with the most pronounced differences in October (+34.22%pt) and December (+34.44%pt).

These patterns reflect the intrinsic nature of both methods. The CAE captures large deviations from learned spatiotemporal patterns, yielding temporally smoother anomaly distributions. In contrast, the z-score method identifies sharp, pixel-wise deviations from the local statistical mean. As a result, the z-score primarily detects abrupt and localized changes, whereas the CAE is sensitive to broader, temporally persistent alterations in SIF. This suggests that the CAE may detect regional or long-term reductions in canopy photosynthetic activity, such as those driven by prolonged drought stress or delayed ecosystem recovery, that are not sufficiently abrupt to be captured by the z-score approach.

To further examine these differences, we focus on different case-study months: September, representing the peak of drought conditions according to both MSB and SPEIbase; May, identified by MSB as the least affected month; and January, identified by SPEIbase as the least affected. Additionally, October and December are examined as months with the largest inter method discrepancies (+34.22%pt and +34.44%pt, respectively). The comparison between the detected SIF anomalies with SPEI and MSB is presented in Sections 3.2 and 3.3, respectively. It is important to note that we do not expect exact overlap as SIF might be affected in a more complex way by droughts than one to one alignment.

Table 1: Fraction of pixels within the study area where TROPOSIF values are considered anomalous (positive or negative) for 2024 based on the convolutional autoencoder (CAE) and z-score methods, expressed as a percentage. The rightmost column shows the difference between these percentage expressed in percentage points.

Month	CAE-based anomalies (%)	Z-score-based anomalies (%)	Anomaly difference (CAE – z-score, %pt)
January	25.82	9.161	16.66
February	17.49	2.674	14.82
March	21.20	6.837	14.36
April	16.18	3.161	13.02
May	26.31	3.151	23.16
June	27.04	0.953	26.09
July	25.26	0.613	24.65
August	34.67	16.279	18.39
September	46.69	35.807	10.88
October	37.95	3.734	34.22
November	26.16	3.763	22.40
December	35.85	1.410	34.44

3.2 Comparison between SIF Anomalies and SPEI

For all months in 2024, we analyzed where $|\text{SPEI}|$ in the study regions exceeded the threshold of 1. As shown in Figure 3, the SPEIbase index identifies extensive areas within the study region affected by dry conditions during 2024. The most affected month is September, with nearly all pixels (94%) classified as dry. The least affected month is January (48.6%), while the second half of the year (June to December) SPEI exhibits substantial drought coverage. This indicates a prolonged drought, with gradual but incomplete recovery from peak drought conditions.

Table 2 reports the Dice and Intersection-over-Union (IoU) coefficients between CAE-detected SIF anomalies and SPEI-based dry conditions. The highest overlap occurs in September (Dice = 0.58, IoU = 0.41), coinciding with the peak drought period. The

lowest value is observed in January (Dice = 0.24), consistent with low drought intensity. This suggests that, under milder drought conditions, the CAE may capture SIF variability influenced by other environmental factors beyond water stress. The overlap remains moderate from October to December, particularly in October (Dice = 0.47). Considering that these months exhibit some of the highest CAE anomaly percentages, the sustained overlap likely reflects prolonged drought stress and delayed ecosystem recovery following the severe dry conditions of September and their milder continuation into the final three months. This suggests that CAE is capturing biological memory or SIF lagged responses.

In contrast, the z-score-based anomalies exhibit substantially lower overlap with SPEI droughts (Table 2). The only notable coincidence occurs in September (Dice = 0.49), when drought severity is at its maximum. For most months, Dice coefficients remain below 0.1, indicating minimal spatial correspondence between z-score anomalies and areas classified as dry by SPEI. There is almost no overlap in months with SPEI anomaly percentages such as June (Dice = 0.03), as well as no overlap in later months where SIF is expected to be low after a prolonged drought such as October (Dice = 0.06).

Overall, the CAE-derived anomalies demonstrate a stronger spatial correspondence with drought conditions identified by SPEI than the z-score method. The z-score approach appears primarily sensitive to short-term, high-intensity events, detecting anomalies only under the most severe drought conditions (e.g., September). In contrast, the CAE captures a broader range of drought-related SIF reductions, including those associated with prolonged stress or delayed vegetation recovery following major drought episodes. This behavior likely arises from the CAE's capacity to model spatiotemporal dependencies, enabling it to recognize persistent deviations in ecosystem function beyond immediate statistical extremes.

3.3 Comparison between SIF Anomalies and MSB

The MSB dataset only contains data for Brazil and we therefore restrict our analysis here to only the Brazilian part of the study area. Binary drought masks were derived from MSB static images by classifying all areas labeled as severe drought ("dark orange" in the MSB color scale) or higher as extreme drought events (value 1), and all lower categories as non-extreme (value 0) as seen in Figure 3.

Table 2 summarizes the overlap between CAE-derived SIF anomalies and MSB drought regions. The highest alignment occurs in September (Dice = 0.85, IoU = 0.45), coinciding with the peak drought period. The lowest score appears in May (Dice = 0.36), corresponding to a month of weaker drought intensity. Notably, October to December maintain moderate to high overlap despite milder drought conditions, suggesting that CAE-detected SIF anomalies remain sensitive to moderate drought stress and may also reflect lingering drought effects and delayed vegetation recovery.

Table 2 shows the corresponding metrics for the z-score method. Here, the intersection with MSB drought areas is substantially lower across most months, except for September (Dice = 0.74, IoU = 0.40). In months with milder or receding droughts, overlap remains minimal (Dice < 0.3), suggesting that SIF z-score anomalies primarily emerge under severe drought intensity and do not persist once conditions begin to improve.

Overall, both methods captured the peak drought conditions in September, but again the CAE consistently demonstrated stronger spatial agreement with MSB drought maps throughout the season, although the months with lighter droughts tend to align less with detected SIF anomalies. The z-score method was more sensitive to short-term, high-intensity deviations, while the CAE detected longer-term, spatially coherent stress signals, likely reflecting the vegetation’s lagged response to prolonged drought and incomplete ecosystem recovery during late 2024.

3.4 Implications of Selected Threshold for Anomaly Detection

Finally, we investigate how the selected anomaly detection threshold ($|z| > 2$ for the z-score method, and a reconstruction error on the 0.99 quantile for the CAE method) influences the percentage of anomalies in both methods. We find that the selected threshold highly impacts the results (see Fig. 4).

Both CAE and the z-score method show a monotonic behavior, where setting higher thresholds results in less pixels being identified as anomalous. A fundamental difference in possible thresholds is that these are bounded at 100% for the CAE method and unbounded for the z-score method. A notable difference in Fig. 4 is that there are ‘jumps’ in the distribution for the CAE method, whereas the distribution for the z-score methods is much smoother. The jumps that we find for the CAE method, illustrate that the

Table 2: Dice coefficient and Intersection over Union (IoU) scores for TROPISIF anomaly detection with both Convolutional Autoencoder (CAE) and z-score methods across months in 2024 against SPEIbase and MSB datasets.

Month	SPEIbase				MSB			
	CAE		z-scores		CAE		z-scores	
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
January	0.2449	0.1395	0.1147	0.0608	0.5039	0.2357	0.2686	0.1389
May	0.2825	0.1645	0.0286	0.0145	0.3552	0.1381	0.0126	0.0058
August	0.4246	0.2695	0.2607	0.1499	0.6662	0.3267	0.4833	0.2737
September	0.5842	0.4126	0.4904	0.3249	0.8511	0.4507	0.7419	0.4006
October	0.4724	0.3093	0.0554	0.0285	0.7348	0.3760	0.0834	0.0408
November	0.3161	0.1877	0.0304	0.0154	0.5173	0.2445	0.1162	0.0588
December	0.3494	0.2117	0.0161	0.0081	0.6487	0.3033	0.0412	0.0206

sensitivity towards the selected threshold can locally be very large, as is also the case close to 0% and close to 100%.

4 Discussion

This study illustrates how different anomaly-detection strategies can lead to markedly different interpretations of vegetation stress in the Amazon. This highlights the non triviality when selecting methodologies for analyzing anomalies in Earth Observation datasets. For instance, the z-score method identifies statistically extreme deviations in pixel-wise time series, emphasizing abrupt and localized changes. This makes it sensitive to sensor artifacts and limits its correspondence with drought conditions within our study area over 2024.

In contrast, the CAE first learns spatiotemporal patterns and then detects deviations from these learned representations. Its smoother month-to-month variation and stronger overlap with both SPEI and MSB suggest that it detects signals related to sustained reductions in canopy photosynthesis and delayed ecosystem recovery. Nevertheless, the CAE requires longer computational time, training effort and expertise (in the

current experiment the training time for CAE on a standard laptop was in the order of 40 minutes). The computational time and associated footprint should also be considered, as is increasingly being recognized (Schwartz et al., 2020).

Ultimately, methodological choices should be guided by the specific research question and the type of anomaly under investigation, rather than convenience or convention. Future work could enhance the use of different methods for anomaly detection of SIF or other vegetation indexes, such as using 3D CAE, Kernel methods (Johnson et al., 2020) or even newly developed foundation models, e.g., DINOv3 (Siméoni et al., 2025) or TerraMind (Jakubik et al., 2025). In Earth Observation, where datasets are high-dimensional and spatiotemporally complex, rigorous consideration of methodological design is essential to ensure that detected anomalies reflect meaningful environmental processes rather than artifacts of the chosen technique.

In addition to the remote sensing data used here, we stress the essential role of in situ measurements for understanding the changing ecosystem dynamics and the associated carbon and water exchange dynamics. In that regard, flux towers in Peru (Griffis et al., 2020), Brazil (Botía et al., 2022; Faassen et al., 2025) and French Guyana (Bonal et al., 2008) are examples of valuable research infrastructures, but overall, the Amazon is relatively poorly sampled (Villarreal & Vargas, 2021). Further investment and expansion in these local measurement stations can provide essential validation for future remote sensing studies.

5 Conclusion

Here we analyzed two different anomaly detection methods, the classical z-score method and the more sophisticated CAE method, applied to SIF data from the Amazon for 2024. We find marked differences between the two methods, with the z-score method showing more abrupt changes and CAE detecting more temporally sustained and spatially coherent anomalies. We stress that the selection of an anomaly detection method deserves more attention in the design of a study as well as in the interpretation of its outcomes.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Open Research Section

In our analysis we used data that is openly available online. The monthly Level-3 TROPISIF data is available at <https://radianteearth.github.io/stac-browser/#/external/data-portal.s5p-pal.com/api/s5p-l3/sif/month>. Drought maps from MSB can be accessed at <https://monitordesecas.ana.gov.br/mapa> and data for SPEIbase is available through <https://spei.csic.es/database.html>. The analysis scripts are available as Supporting Information.

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References

- Agência Nacional de Águas e Saneamento Básico. (2025). *Monitor de Secas: Mapa de Secas*. Retrieved from <https://monitordesecas.ana.gov.br> (Accessed: 2025-06-09)
- Anema, J. C. S., Boersma, K. F., Stammes, P., Koren, G., Woodgate, W., Köhler, P., ... Stol, J. (2024). Monitoring the impact of forest changes on carbon uptake with solar-induced fluorescence measurements from GOME-2A and TROPOMI for an Australian and Chinese case study. *Biogeosciences*, 21(9), 2297–2311. doi: 10.5194/bg-21-2297-2024
- Aragão, L. E. O. C., Anderson, L. O., Fonseca, M. G., Rosan, T. M., Vedovato, L. B., Wagner, F. H., ... Saatchi, S. (2018, Feb 13). 21st Century drought-related fires counteract the decline of Amazon deforestation carbon emissions. *Nature Communications*, 9(1), 536. doi: 10.1038/s41467-017-02771-y
- Beguería, S., Vicente-Serrano, S. M., & Angulo-Martínez, M. (2010). A Multi-scalar Global Drought Dataset: The SPEIbase: A New Gridded Product for the Analysis of Drought Variability and Impacts. *Bulletin of the American Meteorological Society*, 91(10), 1351 - 1356. doi: 10.1175/2010BAMS2988.1
- Bonal, D., Bosc, A., Ponton, S., Goret, J.-Y., Burban, B., Gross, P., ... Granier, A. (2008). Impact of severe dry season on net ecosystem exchange in the Neotropical rainforest of French Guiana. *Global Change Biology*, 14(8), 1917-1933. doi:

- 10.1111/j.1365-2486.2008.01610.x
- Botía, S., Munassar, S., Koch, T., Custodio, D., Basso, L. S., Komiya, S., ... Gerbig, C. (2025). Combined CO₂ measurement record indicates Amazon forest carbon uptake is offset by savanna carbon release. *Atmospheric Chemistry and Physics*, 25(12), 6219–6255. doi: 10.5194/acp-25-6219-2025
- Botía, S., Dias-Junior, C. Q., Komiya, S., Woude, A. v. d., Terristi, M., Kok, R. d., ... Bastos, A. (2025, January). Reduced vegetation uptake during the extreme 2023 drought turns the Amazon into a weak carbon source. *ESS Open Archive*. doi: 10.22541/essoar.173757133.37080321/v1
- Botía, S., Komiya, S., Marshall, J., Koch, T., Gałkowski, M., Lavric, J., ... Gerbig, C. (2022). The CO₂ record at the Amazon Tall Tower Observatory: A new opportunity to study processes on seasonal and inter-annual scales. *Global Change Biology*, 28(2), 588–611. doi: 10.1111/gcb.15905
- Brienen, R. J. W., Phillips, O. L., Feldpausch, T. R., Gloor, E., Baker, T. R., Lloyd, J., ... Zagt, R. J. (2015, Mar 01). Long-term decline of the Amazon carbon sink. *Nature*, 519(7543), 344–348. doi: 10.1038/nature14283
- Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., ... Kohli, P. (2025). *AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data*.
- Buitink, J., Swank, A. M., van der Ploeg, M., Smith, N. E., Benninga, H.-J. F., van der Bolt, F., ... Teuling, A. J. (2020). Anatomy of the 2018 agricultural drought in the Netherlands using in situ soil moisture and satellite vegetation indices. *Hydrology and Earth System Sciences*, 24(12), 6021–6031. doi: 10.5194/hess-24-6021-2020
- Chambers, J. Q., Nogueira Lima, A. J., Pastorello, G., Gimenez, B. O., Meng, L., Dyer, L. A., ... Higuchi, N. (2026, Jan 01). Hot droughts in the Amazon provide a window to a future hypertropical climate. *Nature*, 649(8099), 1190–1196. doi: 10.1038/s41586-025-09728-y
- Dice, L. R. (1945). Measures of the Amount of Ecologic Association Between Species. *Ecology*, 26(3), 297–302. doi: 10.2307/1932409
- Doughty, R., Köhler, P., Frankenberg, C., Magney, T. S., Xiao, X., Qin, Y., ... Moore, B. (2019). TROPOMI reveals dry-season increase of solar-induced chlorophyll fluorescence in the Amazon forest. *Proceedings of the National*

- 449 *Academy of Sciences*, 116(44), 22393-22398. doi: 10.1073/pnas.1908157116
- 450 Espinoza, J.-C., Jimenez, J. C., Marengo, J. A., Schongart, J., Ronchail, J., Lavado-
 451 Casimiro, W., & Ribeiro, J. V. M. (2024, Apr 06). The new record of drought
 452 and warmth in the Amazon in 2023 related to regional and global climatic
 453 features. *Scientific Reports*, 14(1), 8107. doi: 10.1038/s41598-024-58782-5
- 454 Faassen, K. A. P., González-Armas, R., Koren, G., Adnew, G. A., van Asperen,
 455 H., de Boer, H., ... Vilà-Guerau de Arellano, J. (2025). Tracing Diurnal
 456 Variations of Atmospheric CO₂, O₂, and $\delta^{13}\text{CO}_2$ Over a Tropical and a
 457 Temperate Forest. *Geophysical Research Letters*, 52(20), e2025GL118016.
 458 (e2025GL118016 2025GL118016) doi: 10.1029/2025GL118016
- 459 Fleischmann, A. S., Papa, F., Hamilton, S. K., Melack, J., Forsberg, B., Val, A., ...
 460 Marmontel, M. (2025). Extreme warming of Amazon waters in a changing
 461 climate. *Science*, 390(6773), 606-611. doi: 10.1126/science.adr4029
- 462 Flores, B. M., Montoya, E., Sakschewski, B., Nascimento, N., Staal, A., Betts, R. A.,
 463 ... others (2024). Critical transitions in the Amazon forest system. *Nature*,
 464 626(7999), 555-564.
- 465 Frankenberg, C., & Berry, J. (2018). 3.10 - Solar Induced Chlorophyll Fluorescence:
 466 Origins, Relation to Photosynthesis and Retrieval. In S. Liang (Ed.), *Compre-
 467 hensive remote sensing* (p. 143-162). Oxford: Elsevier. doi: 10.1016/B978-0-12
 468 -409548-9.10632-3
- 469 Gatti, L. V., Basso, L. S., Miller, J. B., Gloor, M., Gatti Domingues, L., Cassol,
 470 H. L., ... others (2021). Amazonia as a carbon source linked to deforestation
 471 and climate change. *Nature*, 595(7867), 388-393.
- 472 Griffis, T., Roman, D., Wood, J., Deventer, J., Fachin, L., Rengifo, J., ... Ricciuto,
 473 D. (2020). Hydrometeorological sensitivities of net ecosystem carbon dioxide
 474 and methane exchange of an Amazonian palm swamp peatland. *Agricultural
 475 and Forest Meteorology*, 295, 108167. doi: 10.1016/j.agrformet.2020.108167
- 476 Guanter, L., Bacour, C., Schneider, A., Aben, I., van Kempen, T. A., Maignan, F.,
 477 ... Zhang, Y. (2021). The TROPOSIF global sun-induced fluorescence dataset
 478 from the Sentinel-5P TROPOMI mission. *Earth System Science Data*, 13(11),
 479 5423-5440. doi: 10.5194/essd-13-5423-2021
- 480 Hubau, W., Lewis, S. L., Phillips, O. L., Affum-Baffoe, K., Beeckman, H., Cuní-
 481 Sanchez, A., ... Zemagho, L. (2020, Mar 01). Asynchronous carbon sink

- saturation in African and Amazonian tropical forests. *Nature*, 579(7797), 80-87. doi: 10.1038/s41586-020-2035-0
- Jakubik, J., Yang, F., Blumenstiel, B., Scheurer, E., Sedona, R., Maurogiovanni, S., ... Longép , N. (2025). *TerraMind: Large-Scale Generative Multimodality for Earth Observation*.
- Jiang, Y., Shi, H., Yang, X., Wen, Z., Wu, Y., Wang, Y., ... Li, J. (2025). Tracking drought dynamics on China's Loess Plateau with the rapid change index of solar-induced chlorophyll fluorescence. *International Journal of Digital Earth*, 18(1), 2473128. doi: 10.1080/17538947.2025.2473128
- Johnson, J. E., Laparra, V., P rez-Suay, A., Mahecha, M. D., & Camps-Valls, G. (2020). Kernel methods and their derivatives: Concept and perspectives for the earth system sciences. *Plos one*, 15(10), e0235885.
- Kooistra, L., Berger, K., Brede, B., Graf, L. V., Aasen, H., Roujean, J.-L., ... Verrelst, J. (2024). Reviews and syntheses: Remotely sensed optical time series for monitoring vegetation productivity. *Biogeosciences*, 21(2), 473–511. doi: 10.5194/bg-21-473-2024
- Koren, G., van Schaik, E., Ara jo, A. C., Boersma, K. F., G rtner, A., Killaars, L., ... Peters, W. (2018). Widespread reduction in sun-induced fluorescence from the Amazon during the 2015/2016 El Ni o. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1760), 20170408. doi: 10.1098/rstb.2017.0408
- Lee, J.-E., Frankenberg, C., van der Tol, C., Berry, J. A., Guanter, L., Boyce, C. K., ... Saatchi, S. (2013, 06). Forest productivity and water stress in Amazonia: observations from GOSAT chlorophyll fluorescence. *Proceedings of the Royal Society B: Biological Sciences*, 280(1761), 20130171. doi: 10.1098/rspb.2013.0171
- Liu, J., Bowman, K. W., Schimel, D. S., Parazoo, N. C., Jiang, Z., Lee, M., ... Eldering, A. (2017). Contrasting carbon cycle responses of the tropical continents to the 2015–2016 El Ni o. *Science*, 358(6360), eaam5690. doi: 10.1126/science.aam5690
- Lu, M., Sun, H., Cheng, L., Li, S., Qin, H., Yi, S., ... Zhang, W. (2024, jun). Heterogeneity in vegetation recovery rates post-flash droughts across different ecosystems. *Environmental Research Letters*, 19(7), 074028. doi:

- 10.1088/1748-9326/ad5570
- Marengo, J. A., Costa, M. C., Cunha, A. P., Espinoza, J.-C., Jimenez, J. C., Libonati, R., ... Medeiros, O. (2026). Characterisation of the Exceptional Heatwave Conditions Observed in Brazil During the Record-Hot Years of 2024 and 2025. *International Journal of Climatology*, e70219. doi: 10.1002/joc.70219
- Marengo, J. A., Souza, C. M., Thonicke, K., Burton, C., Halladay, K., Betts, R. A., ... Soares, W. R. (2018). Changes in climate and land use over the amazon region: Current and future variability and trends. *Frontiers in Earth Science, Volume 6 - 2018*. doi: 10.3389/feart.2018.00228
- Mengistu, A. G., Mengistu Tsidu, G., Koren, G., Kooreman, M. L., Boersma, K. F., Tagesson, T., ... Peters, W. (2021). Sun-induced fluorescence and near-infrared reflectance of vegetation track the seasonal dynamics of gross primary production over Africa. *Biogeosciences*, 18(9), 2843–2857. doi: 10.5194/bg-18-2843-2021
- Naus, S., Domingues, L. G., Krol, M., Luijkx, I. T., Gatti, L. V., Miller, J. B., ... Peters, W. (2022). Sixteen years of MOPITT satellite data strongly constrain Amazon CO fire emissions. *Atmospheric Chemistry and Physics*, 22(22), 14735–14750. doi: 10.5194/acp-22-14735-2022
- Rosan, T. M., Sitch, S., O’Sullivan, M., Basso, L. S., Wilson, C., Silva, C., ... Aragão, L. E. O. C. (2024, Jan 22). Synthesis of the land carbon fluxes of the Amazon region between 2010 and 2020. *Communications Earth & Environment*, 5(1), 46. doi: 10.1038/s43247-024-01205-0
- Schwartz, R., Dodge, J., Smith, N. A., & Etzioni, O. (2020). Green AI. *Communications of the ACM*, 63(12), 54–63. doi: 10.1145/3381831
- Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., ... others (2025). Dinov3. *arXiv preprint arXiv:2508.10104*.
- Smallman, T. L., Milodowski, D. T., Neto, E. S., Koren, G., Ometto, J., & Williams, M. (2021). Parameter uncertainty dominates C-cycle forecast errors over most of Brazil for the 21st century. *Earth System Dynamics*, 12(4), 1191–1237. doi: 10.5194/esd-12-1191-2021
- Smith, N. E., Kooijmans, L. M. J., Koren, G., van Schaik, E., van der Woude, A. M., Wanders, N., ... Peters, W. (2020, 09). Spring enhancement and summer reduction in carbon uptake during the 2018 drought in northwestern

- Europe. *Philosophical Transactions of the Royal Society B: Biological Sciences*,
375(1810), 20190509. doi: 10.1098/rstb.2019.0509
- Staal, A., Koren, G., Tejada, G., & Gatti, L. V. (2023, apr). Moisture origins of the
Amazon carbon source region. *Environmental Research Letters*, 18(4), 044027.
doi: 10.1088/1748-9326/acc676
- Sun, Z., ten Brink, T., Carande, W., Koren, G., Cristea, N., Jorgenson, C.,
... Duzgun, S. (2024, Dec 01). Towards practical artificial intelligence
in Earth sciences. *Computational Geosciences*, 28(6), 1305-1329. doi:
10.1007/s10596-024-10317-7
- Toreti, A., Bavera, D., Acosta Navarro, J., Acquafresca, L., Barbosa, P., De Jager,
A., ... Volpi, D. (2024). *Drought in the Amazon and the La Plata Basins –
December 2024 – GDO analytical report*. Publications Office of the European
Union. doi: doi/10.2760/9524487
- Turner, A. J., Köhler, P., Magney, T. S., Frankenberg, C., Fung, I., & Cohen,
R. C. (2020). A double peak in the seasonality of california’s photo-
synthesis as observed from space. *Biogeosciences*, 17(2), 405–422. doi:
10.5194/bg-17-405-2020
- Van Passel, J., Bernardino, P. N., Lhermitte, S., Rius, B. F., Hirota, M., Conradi,
T., ... Somers, B. (2024). Critical slowing down of the Amazon forest after in-
creased drought occurrence. *Proceedings of the National Academy of Sciences*,
121(22), e2316924121. doi: 10.1073/pnas.2316924121
- van Schaik, E., Killaars, L., Smith, N. E., Koren, G., van Beek, L. P. H., Peters, W.,
& van der Laan-Luijkx, I. T. (2018, 10). Changes in surface hydrology, soil
moisture and gross primary production in the Amazon during the 2015/2016
El Niño. *Philosophical Transactions of the Royal Society B: Biological Sci-
ences*, 373(1760), 20180084. doi: 10.1098/rstb.2018.0084
- Veefkind, J., Aben, I., McMullan, K., Förster, H., de Vries, J., Otter, G., ... Lev-
elt, P. (2012). TROPOMI on the ESA Sentinel-5 Precursor: A GMES
mission for global observations of the atmospheric composition for climate,
air quality and ozone layer applications. *Remote Sensing of Environment*,
120, 70-83. (The Sentinel Missions - New Opportunities for Science) doi:
10.1016/j.rse.2011.09.027
- Vicente-Serrano, S. M., Beguería, S., & López-Moreno, J. I. (2010). A Multiscalar

- 581 Drought Index Sensitive to Global Warming: The Standardized Precipita-
 582 tion Evapotranspiration Index. *Journal of Climate*, 23(7), 1696 - 1718. doi:
 583 10.1175/2009JCLI2909.1
- 584 Villarreal, S., & Vargas, R. (2021). Representativeness of FLUXNET Sites Across
 585 Latin America. *Journal of Geophysical Research: Biogeosciences*, 126(3),
 586 e2020JG006090. (e2020JG006090 2020JG006090) doi: 10.1029/2020JG006090
- 587 Zhang, Y., & Zhang, Z. (2026). 3.12 - Solar Induced Chlorophyll Fluorescence: Ori-
 588 gins and Applications, Relation to Photosynthesis and Retrieval. In S. Liang
 589 (Ed.), *Comprehensive remote sensing (second edition)* (Second Edition ed.,
 590 p. 257-296). Oxford: Elsevier. doi: 10.1016/B978-0-443-13220-9.00055-X
- 591 Zhang, Z., Cescatti, A., Wang, Y.-P., Gentile, P., Xiao, J., Guanter, L., ... Zhang,
 592 Y. (2023). Large diurnal compensatory effects mitigate the response of Ama-
 593 zonian forests to atmospheric warming and drying. *Science Advances*, 9(21),
 594 eabq4974. doi: 10.1126/sciadv.abq4974
- 595 Zhu, X. X., Tuia, D., Mou, L., Xia, G.-S., Zhang, L., Xu, F., & Fraundorfer, F.
 596 (2017). Deep Learning in Remote Sensing: A Comprehensive Review and List
 597 of Resources. *IEEE Geoscience and Remote Sensing Magazine*, 5(4), 8-36. doi:
 598 10.1109/MGRS.2017.2762307

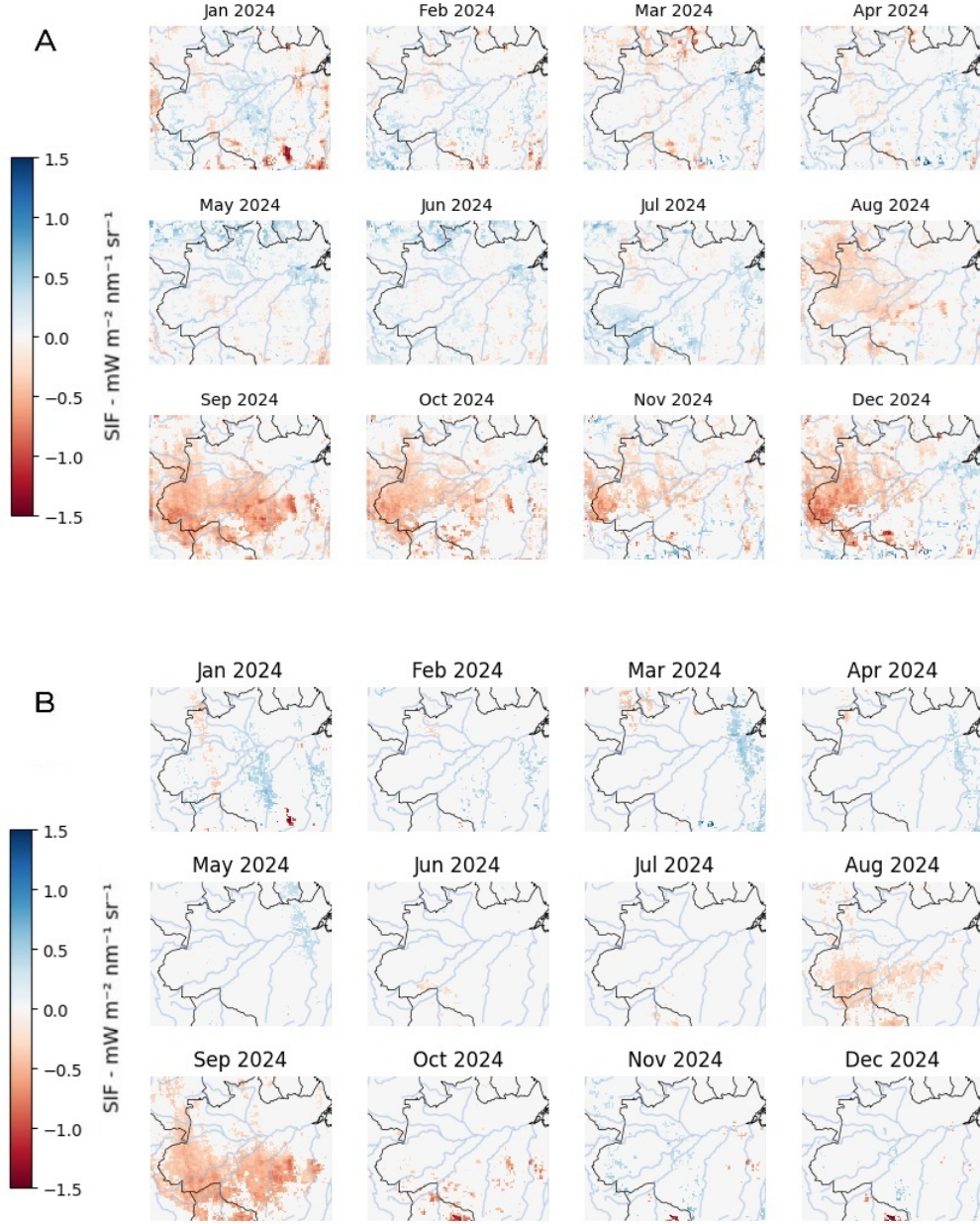


Figure 2: Detected TROPOSIF anomalies over the study area for 2024 using (A) the CAE method and (B) Z-score method. The anomalous pixels (i.e., for the CAE method the pixels where the reconstruction error exceeds the 99th percentile and for the Z-score method the pixels where $|z| > 2$) are colored and other pixels are white. The color corresponds to the size and sign of the anomalies (i.e., for the CAE method the difference between observation and reconstructed image, and for the z-score method the difference between the observation and the mean for that month and pixel over years 2019-2023).

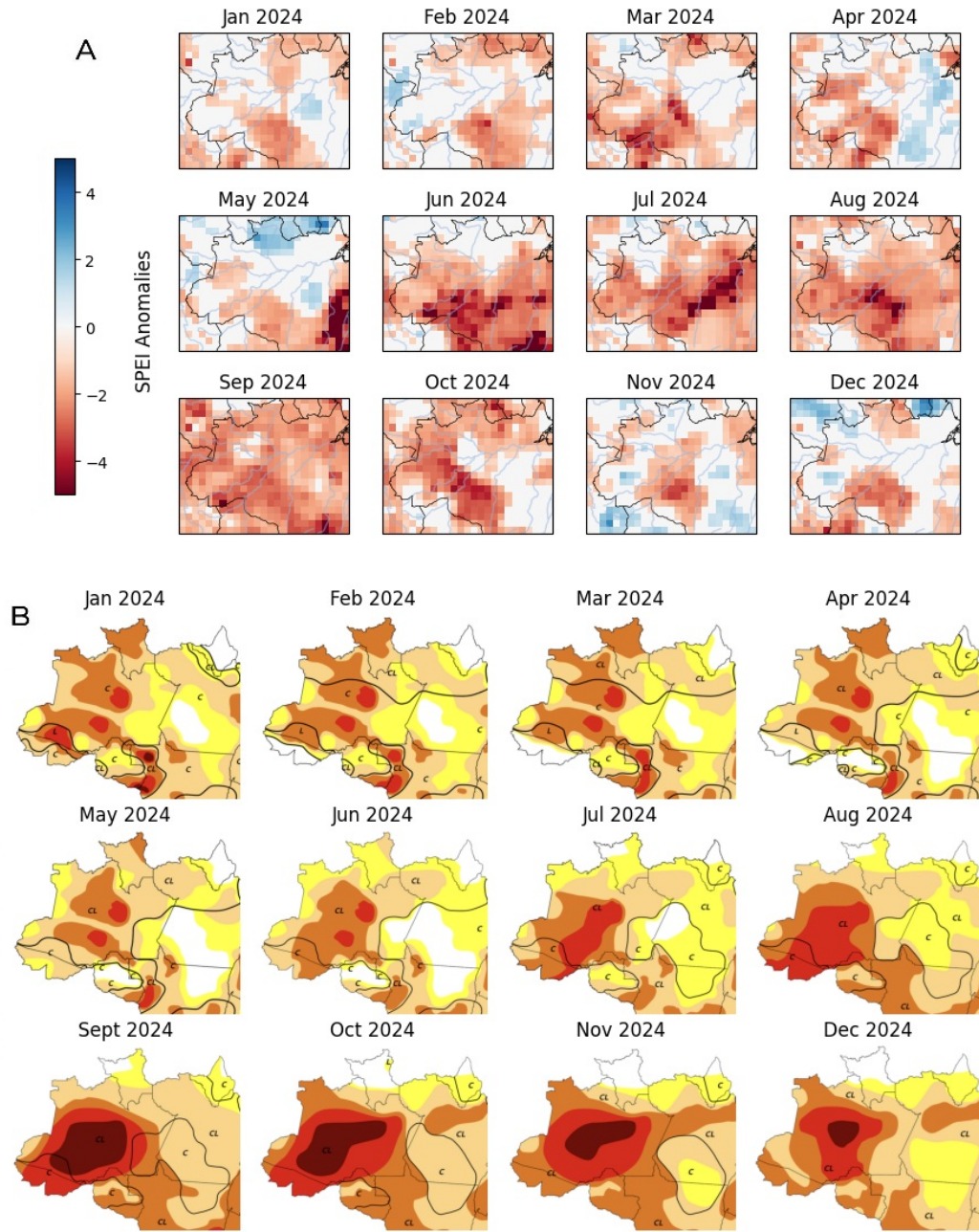


Figure 3: Drought indicator anomalies for months of 2024 (A) Standardized Precipitation Evapotranspiration Index (SPEI) values for cells where $|\text{SPEI}| > 1$. (B) Monitor de Secas do Brasil (MSB) anomalies for Brazilian part of the study area. Yellow is mild drought, light orange is moderate drought, orange is severe drought, light red is extreme drought and dark red is exceptional drought.

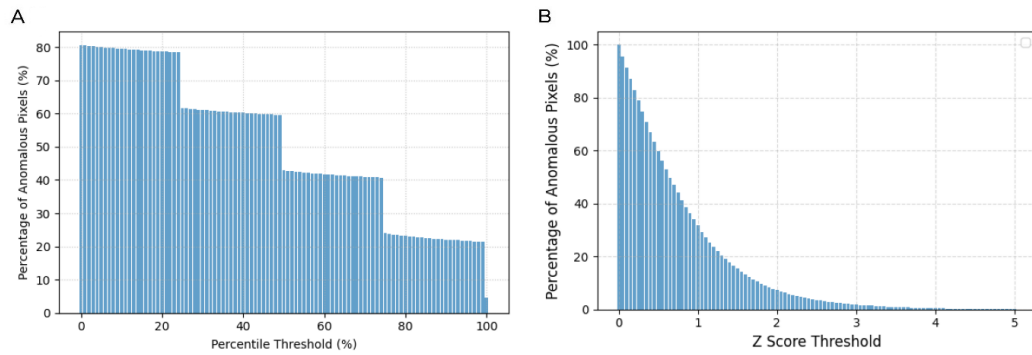


Figure 4: Percentage of pixels that are identified as being anomalous depending on the selected threshold for (A) reconstruction error of the CAE method and (B) z-score for the z-score method.